

CSE36301 Machine Learning Assignment3

Deadline: Monday, November 17, 2025, 11:59 PM

Introduction

In this assignment, you will explore three fundamental topics in machine learning using the **Breast Cancer Wisconsin Dataset** from `scikit-learn`:

1. **Ensemble Learning:** Implement and compare various ensemble methods including Decision Tree, Random Forest, Bagging, and AdaBoost.
2. **Dimension Reduction:** Apply Principal Component Analysis (PCA) and analyze its impact on model performance.
3. **Neural Network:** Build and train a simple Multi-Layer Perceptron (MLP) using PyTorch, and compare its results with ensemble models.

The goal of this assignment is to understand **how different models handle bias–variance trade-offs, the effect of dimensionality on learning, and the stability of algorithms under limited data conditions**. In addition to theoretical understanding, this assignment emphasizes **hands-on implementation**. Students are required to implement several algorithms from scratch (e.g., Decision Tree, Bagging, AdaBoost, PCA) following the provided code skeletons. Each section contains `TODO` comments that guide you step by step through the implementation process. You are encouraged to carefully read the comments, fill in the missing parts, and verify your results through visualization and performance comparison.

1 Ensemble Learning

Dataset: Breast Cancer Wisconsin Dataset (569×30)

Models: Decision Tree (manual implementation), Random Forest, Bagging (manual implementation), AdaBoost (manual implementation)

Goal: Understand how ensemble methods such as Decision Tree, Random Forest, Bagging, and AdaBoost improve model stability and performance through variance reduction and error correction, and analyze their behavior under different data sizes and hyperparameter settings.

T1. Implement a Decision Tree classifier using the Gini impurity as the splitting criterion.

T2. Implement the bootstrap sampling procedure for the Bagging classifier.

T3. Implement the core steps of the AdaBoost algorithm.

- (a) **Compute weighted error:** Calculate the misclassification rate using the current sample weights.
- (b) **Compute model weight (α):** Assign a higher weight to models with lower error using the formula:

$$\alpha = \frac{1}{2} \log \left(\frac{1 - \text{err}}{\text{err} + 10^{-10}} \right)$$

- (c) **Update sample weights:** Increase the weights of misclassified samples and decrease the weights of correctly classified ones, then normalize so that all weights sum to 1.

- T4.** Visualize the feature importances from the Random Forest model using a bar plot.
- T5.** Create a bar plot comparing the accuracy of Decision Tree, Random Forest, Bagging, and Adaboost.
- Q1.** Change the `test_size` hyperparameter and observe which ensemble method remains the most stable when the dataset is relatively small. Experiment by varying `n_estimators`.
- Q2.** Select the 10 most important features from the Random Forest, visualize them, and analyze how the accuracy changes when training with only these features.

2 Dimension Reduction

Goal: Apply PCA to reduce the dataset dimensionality and analyze the trade-off between dimensionality and classification performance.

- T1.** Implement PCA from scratch using NumPy by computing the covariance matrix, performing eigen decomposition, and projecting the data onto the top principal components.
- T2.** Using the given dataset, apply PCA and visualize the results. What is the total retained variance with 2 components? Report and interpret the PCA visualization.
- Q1.** How many principal components are required to retain more than 95% of the variance?
- Q2.** Apply Random Forest after PCA and compare the performance. Set `n_estimators = 100`.
- Q3.** Compare the performance between using the top n features selected by Random Forest feature importance and using n PCA components.

3 Neural Network

Goal: Train a simple feed-forward neural network for binary classification and analyze its learning stability.

- T1.** Define a simple MLP with one hidden layer using ReLU activation function.
- T2.** Implement the forward pass, compute the loss, and perform backpropagation for each training batch.
- Q1.** Change `hidden_dim` to 16, 32, and 64, and find which configuration achieves the best performance.
- Q2.** Change the learning rate (`lr`) to 0.001, 0.01, and 0.1, and compare training stability.
- Q3.** Discuss which model—among Decision Tree, Random Forest, Bagging, AdaBoost, and MLP—is the most suitable for this dataset.

Submission Guidelines

- Your submission must be a `.zip` file containing the following files with the exact naming convention: `Assignment3_[YOUR_STUDENT_ID]_[COMPONENT].[EXTENSION]`
- Submit a Jupyter Notebook (`.ipynb`) including all code, results.
- Submit a written report including your analyses and all relevant plots (accuracy comparison, PCA visualization, learning curves, etc.).
- Code must be well-commented and reproducible.

Evaluation Criteria

- Correctness and completeness of implementation (50%)
- Depth of analysis and reasoning (25%)
- Clarity of visualizations and explanations (25%)

Contact

If you have any questions regarding this assignment, please email the TAs:

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